

PREDICTING CUSTOMER CHURN:

A DATA-DRIVEN ANALYSIS OF SUBSCRIPTION CUSTOMER BEHAVIOUR

Using Classification Models to Identify At-Risk Telecom Customers

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DECLARATION

IDRA Capstone 2026 — Customer Churn

I hereby declare that the work presented in this project report, titled “Predicting Customer Churn: A Data Driven Analysis of Subscription Customer Behaviour,” is my own original work carried out as part of the Data Science and AI Training Program conducted by the India Data Research Academy (IDRA). The data analysis, exploratory analysis, statistical reasoning, and machine learning modelling described in this report, together with the accompanying notebook, were completed by me. All external sources, datasets, tools, and references used during the project have been duly acknowledged. I further declare that this report has not been submitted elsewhere for the award of any other qualification.

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Signature:

A handwritten signature in blue ink, appearing to read 'Inbakumaran S', with a long horizontal stroke extending to the right.

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ABSTRACT

Customer churn — the loss of subscribers to competitors or cancellation — is one of the most costly problems faced by subscription businesses, because retaining an existing customer is generally far cheaper than acquiring a new one. This project investigates the drivers of churn in a telecommunications company and builds a predictive model to identify customers at risk of leaving. The analysis uses the Telco Customer Churn dataset of 7,043 customers described by 19 predictor variables spanning demographics, account information, subscribed services, contract type, and billing behaviour, with a binary churn indicator as the target. After cleaning — most notably converting the TotalCharges field from text to numeric and resolving 11 blank values belonging to newly-joined customers — the data were encoded, split into training and test sets, and scaled without leakage. Exploratory analysis revealed that churn is concentrated among month-to-month contract holders, customers in their first year of tenure, and fibre-optic internet subscribers. A Logistic Regression classifier and a Random Forest classifier were trained and compared. Logistic Regression achieved 80.6% test accuracy with a churn-class F1-score of 0.61 and generalised cleanly, whereas the Random Forest overfitted heavily (99.8% train versus 79.4% test). The results confirm contract length and tenure as the dominant predictors and recommend a targeted early-lifecycle retention programme for high risk segments.

Keywords: Customer churn; Classification; Logistic Regression; Exploratory Data Analysis; Data Preprocessing; Predictive Analytics

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1. INTRODUCTION

1.1 Background

Subscription-based businesses — telecom operators, streaming platforms, software providers and utilities — depend on recurring revenue from a stable customer base. In these industries, customer churn refers to the phenomenon of subscribers discontinuing their service within a given period. Churn matters because acquiring a new customer typically costs several times more than retaining an existing one, and a high churn rate erodes revenue even when a company is winning new customers. Understanding why customers leave, and being able to predict who is likely to leave next, therefore has direct commercial value. This project applies the data science workflow — data cleaning, exploratory analysis, statistics, visualization and machine learning — to a real telecommunications dataset in order to understand and predict churn.

1.2 Context of the Problem

The dataset used here represents customers of a fictional telecommunications company offering phone and internet services. Each record describes a single customer's demographic profile, the services they subscribe to, their contract and payment arrangements, their charges, and whether they left the company in the most recent period. In a real setting, the stakeholders for this analysis would be the company's retention, marketing and customer-experience teams, who need to know both which factors are associated with churn and which individual customers are most at risk, so that limited retention budgets can be directed where they will have the greatest effect.

1.3 Scope of the Study

This study covers the full analytical pipeline on the provided cross-sectional dataset: data quality assessment and cleaning, exploratory and statistical analysis of churn patterns, and the development and evaluation of classification models to predict churn. The study does not attempt to establish causal relationships, does not incorporate external data such as competitor pricing or network-quality measurements, and does not deploy the model into a live system. It is a single-snapshot analysis intended to surface actionable patterns and to demonstrate a reproducible predictive workflow.

2. PROBLEM STATEMENT AND MOTIVATION

The central problem this project addresses can be stated as a single question: which characteristics distinguish customers who churn from those who stay, and can these characteristics be used to predict churn accurately enough to support retention decisions? The motivation is practical and financial. Because retaining a customer is cheaper than replacing one, even a modest improvement in the ability to flag at-risk customers before they leave can translate into meaningful savings. A data-driven solution offers two forms of value: it explains the drivers of churn in interpretable terms that management can act on, and it produces individual-level risk predictions that allow retention efforts to be prioritised rather than applied uniformly.

3. OBJECTIVES

The project pursues the following measurable objectives:

- To clean and prepare the Telco Customer Churn dataset for analysis, resolving all identified data quality issues.
- To identify, through exploratory and statistical analysis, the variables most strongly associated with customer churn.

- To build classification models that predict whether a customer will churn.
- To evaluate the models using accuracy, precision, recall, F1-score and a confusion matrix, and to assess overfitting by comparing training and test performance.
- To translate the findings into specific, evidence-based retention recommendations.

4. RESEARCH QUESTIONS

1. What are the major characteristics and data-quality issues present in the churn dataset?
2. Which variables — such as contract type, tenure, internet service and billing method — are most strongly associated with churn?
3. How does the churn rate vary across important customer groups?
4. How accurately can a classification model predict churn on unseen data, and which model generalises best?

5. DATASET DESCRIPTION AND DATA UNDERSTANDING

5.1 Data Source

The dataset is the Telco Customer Churn dataset supplied through the IDRA Learning Management System for this project (file: P_4_WA_Fn-UseC_-Telco-Customer-Churn.csv). It is a widely-used, openly-available benchmark dataset for churn analysis. No personally identifying information is present beyond an anonymised customer identifier, which was removed during cleaning.

5.2 Dataset Size and Structure

The raw dataset contains 7,043 rows (one per customer) and 21 columns. Of these, one column (customerID) is an identifier, nineteen are candidate predictor variables, and one — Churn — is the target. Churn is a binary categorical variable (Yes/No) indicating whether the customer left in the most recent period, which makes this a classification problem. As summarised in Table 1, the classes are imbalanced: roughly one customer in four churned.

Property	Value
Number of rows (customers)	7,043
Number of columns (raw)	21
Predictor variables	19 (after removing the ID)
Target variable	Churn (Yes / No)
Customers who churned	1,869 (26.5%)
Customers who stayed	5,174 (73.5%)
Problem type	Binary classification

Table 1. Summary of the Dataset

5.3 Data Dictionary

Table 2 describes the main variables used in the analysis, grouped by their role.

Variable	Meaning	Type	Role
gender	Customer gender	Categorical	Feature

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Variable	Meaning	Type	Role
SeniorCitizen	Whether the customer is a senior citizen	Categorical (0/1)	Feature
Partner / Dependents	Whether the customer has a partner / dependents	Categorical	Feature
tenure	Number of months the customer has stayed	Numeric	Feature
PhoneService / MultipleLines	Phone-service subscriptions	Categorical	Feature
InternetService	DSL, Fibre optic, or None	Categorical	Feature
OnlineSecurity ... StreamingMovies	Add-on internet services	Categorical	Feature
Contract	Month-to-month, One year, Two year	Categorical	Feature
PaperlessBilling / PaymentMethod	Billing preferences	Categorical	Feature
MonthlyCharges	Amount charged monthly	Numeric	Feature
TotalCharges	Total amount charged to date	Numeric	Feature
Churn	Whether the customer left	Categorical	Target

Table 2. Data Dictionary

6. DATA CLEANING

Before analysis, the raw data were assessed for the standard quality issues: missing values, duplicates, incorrect data types, inconsistent categories and outliers. Each decision below is recorded with its reason, and the effect of cleaning is validated at the end of the section.

6.1 Missing Values

On loading, no column reported missing values, but the TotalCharges column — which represents money — had been read as text. Converting it to a numeric type revealed 11 entries (0.16% of all rows) that were

actually blank. Inspecting these rows showed that every one belonged to a customer with a tenure of 0 months — that is, brand-new customers who had not yet been billed for a full cycle. Because a total charge of zero is the logically correct value for a customer who has just joined, these 11 blanks were filled with 0 rather than dropped. Dropping them would have discarded valid customers and mildly biased the sample against new joiners, which the project guide explicitly cautions against.

6.2 Duplicate Records

The dataset was checked for fully duplicated rows and for duplicated customer identifiers. None were found, so no rows were removed on this basis and the dataset size was unchanged.

6.3 Data Types and Inconsistent Values

Two type corrections were applied. First, TotalCharges was converted from text to a floating-point number, as described above. Second, SeniorCitizen was stored as an integer (0/1) but is conceptually a Yes/No category; it was relabelled to “No”/“Yes” so that it would be treated correctly during exploratory analysis and encoding. The customerID column was removed entirely, since a unique identifier carries no predictive information and could only introduce noise. Category labels in the service columns (for example, “No internet service”) were consistent and required no correction.

6.4 Outlier Handling

The three numeric variables — tenure, MonthlyCharges and TotalCharges — were examined for outliers. Their ranges are all commercially plausible (tenure spans 0–72 months, MonthlyCharges roughly 18 to 119, and TotalCharges up to about 8,685). The high TotalCharges values are not errors but simply long-standing, high-spending customers, so no observations were removed or capped. Retaining them preserves genuine signal about the customer base.

6.5 Before vs After Cleaning

Check	Before cleaning	After cleaning
Rows	7,043	7,043
TotalCharges data type	Text (object)	Numeric (float)
Missing / blank values	11 (in TotalCharges)	0
Duplicate rows	0	0
Identifier column	Present (customerID)	Removed
SeniorCitizen encoding	Integer 0/1	Categorical No/Yes

Table 3. Data Quality Before and After Cleaning

7. DATA PREPROCESSING

After cleaning, the dataset was prepared for modelling. The steps below distinguish clearly between operations performed on the whole dataset and those fitted only on the training data to avoid leakage.

7.1 Categorical Encoding

The binary target Churn was mapped to 1 (Yes) and 0 (No). The categorical predictor variables were converted to numeric form using one-hot encoding (pandas `get_dummies` with `drop_first=True`). One-hot encoding was chosen because most of these variables are nominal — they have no natural order — so label encoding would wrongly imply a ranking. Dropping the first category of each variable avoids redundant, perfectly-correlated columns. Encoding expanded the feature set to 30 numeric columns.

7.2 Feature Scaling and Transformation

The three continuous variables — tenure, MonthlyCharges and TotalCharges — were standardised (zero mean, unit variance) using a `StandardScaler`. Scaling is important for Logistic Regression, whose coefficients are sensitive to feature magnitude, so that no single large-valued feature dominates purely because of its units. The one-hot columns, already on a 0/1 scale, were left unchanged.

7.3 Train-Test Split

The data were split into 80% training and 20% test sets (5,634 and 1,409 customers respectively) with a fixed random seed of 42 for reproducibility. The split used stratification on the target so that the 26.5% churn proportion is preserved in both sets — important for imbalanced data. Crucially, the `StandardScaler` was fitted on the training data only and then applied to the test data. This ordering prevents information from the test set influencing the transformation, which would otherwise cause data leakage and optimistic results.

7.4 Final Feature Set

The final model input consisted of 30 encoded and scaled features derived from the customer's demographics, subscribed services, contract type, billing method and charges. The target variable was the binary Churn indicator. No columns that reveal the outcome after the fact were included, so there is no obvious target leakage.

8. EXPLORATORY DATA ANALYSIS

The exploratory analysis focused on the questions defined in Section 4. Only charts that support a finding are included here; the interpretation follows each figure.

8.1 Univariate Analysis

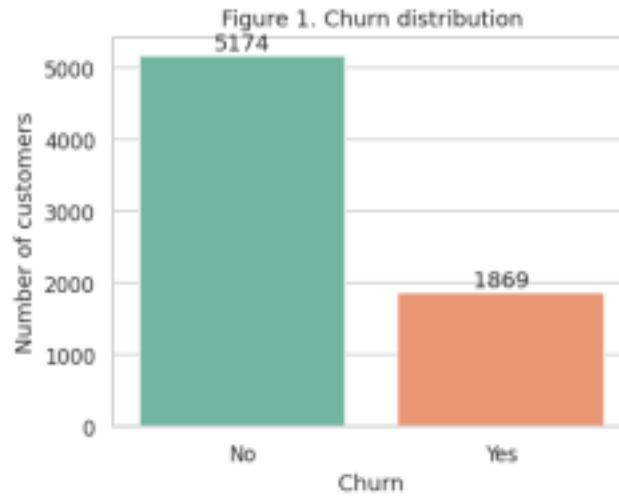


Figure 1. Distribution of the target variable (Churn)

Figure 1 shows that the target is imbalanced: 5,174 customers stayed and 1,869 churned, so churners make up about 26.5% of the sample. This imbalance is important for later evaluation — a naive model that always predicts “no churn” would already be about 73% accurate while being useless for retention. It is the reason precision, recall and F1-score are emphasised over raw accuracy.

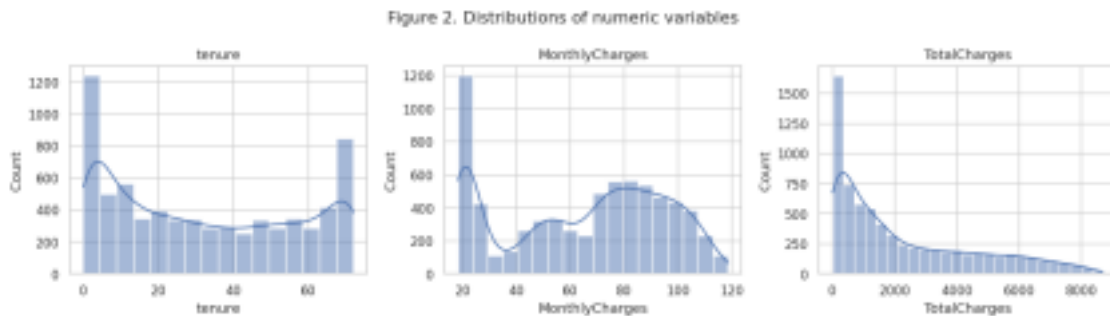


Figure 2. Distributions of tenure, MonthlyCharges and TotalCharges

Figure 2 shows that tenure and TotalCharges are right-skewed, with a large concentration of customers at low values — many customers are relatively new. MonthlyCharges is broadly spread with a cluster of low cost (likely phone-only) customers and a second concentration at higher charges. Because these distributions are skewed, the median is a more representative measure of centre than the mean, as reflected in the statistics in Section 9.

8.2 Categorical Analysis

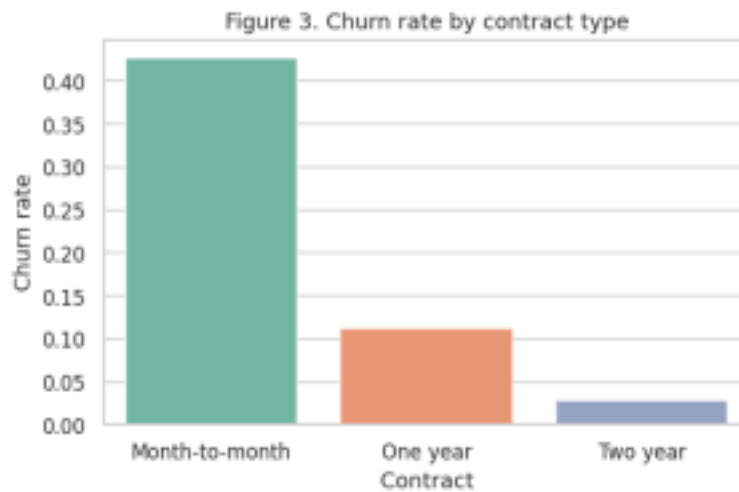


Figure 3. Churn rate by contract type

Figure 3 reveals a striking pattern: month-to-month customers churn at 42.7%, compared with just 11.3% for one-year contracts and 2.8% for two-year contracts. Contract length is therefore one of the strongest signals in the entire dataset — the commitment implied by a longer contract is closely tied to retention.

8.3 Bivariate / Relationship Analysis

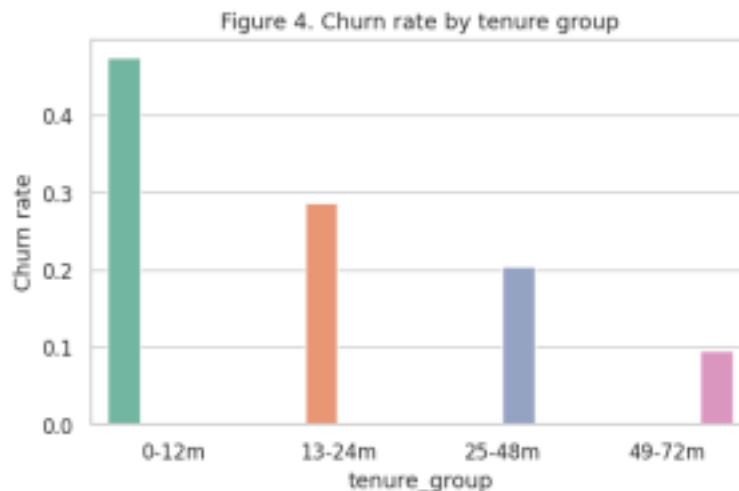


Figure 4. Churn rate by tenure group

Figure 4 shows churn falling sharply as tenure increases: 47.4% of customers in their first year churn, dropping to 28.7% in the second year, 20.4% between two and four years, and only 9.5% beyond four years. The relationship is clear and monotonic — the earliest months of the customer lifecycle carry by far the highest risk. This is a relationship rather than proof of causation, but it strongly suggests where retention effort should be concentrated.

8.4 Correlation Analysis

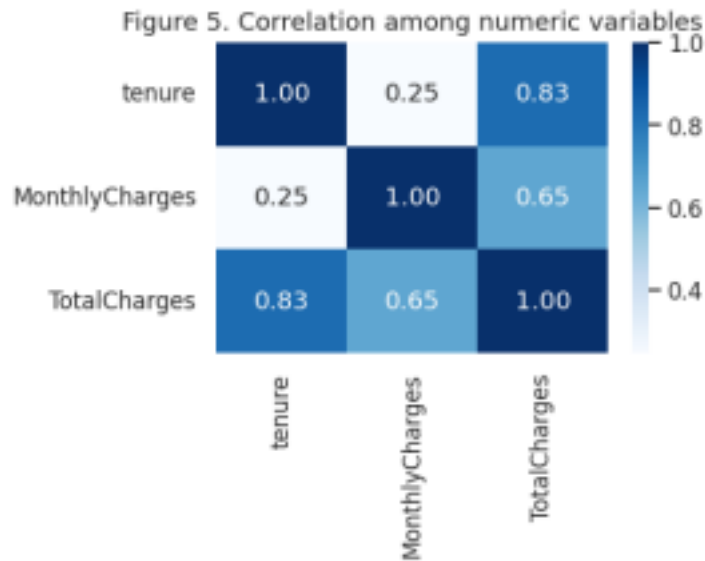


Figure 5. Correlation heatmap of numerical variables

Figure 5 shows a strong positive correlation between tenure and TotalCharges, which is expected because a customer who stays longer accumulates more total charges over time. MonthlyCharges correlates moderately with TotalCharges but weakly with tenure. These relationships are logical rather than surprising, and they warn against treating tenure and TotalCharges as fully independent signals in interpretation.

9. STATISTICAL ANALYSIS

Table 4 reports descriptive statistics for the numeric variables. The gap between the mean and median for tenure and TotalCharges confirms the right-skew seen in Figure 2.

Statistic	tenure (months)	MonthlyCharges	TotalCharges
Mean	32.37	64.76	2,279.73
Median	29.00	70.35	1,394.55
Std. deviation	24.56	30.09	2,266.79
Minimum	0.00	18.25	0.00
Maximum	72.00	118.75	8,684.80
IQR	46.00	54.35	3,388.05

Table 4. Descriptive Statistics of Key Variables

The statistics reinforce the exploratory findings. For TotalCharges the mean (2,279.73) sits well above the median (1,394.55), a signature of right-skew driven by a minority of long-tenured, high-spending customers; the median is therefore the more representative centre. Tenure shows the same pattern more mildly. Group-level churn summaries computed during EDA — 42.7% churn for month-to-month contracts, 41.9% for fibre-optic internet, 45.3% for electronic-check payers, and 47.4% for first-year customers — quantify the segments most exposed to churn and directly inform the recommendations.

10. FEATURE ENGINEERING

The project relied primarily on the variables already present in the dataset, transformed through encoding and scaling. One interpretive feature, a tenure group, was created during exploratory analysis to make the tenure–churn relationship easier to read (Figure 4); it was used for visualization only and excluded from the

model to avoid duplicating the information already carried by the continuous tenure variable. Table 5 summarises the engineered and derived features.

Feature	Derived from	Purpose	Used in model?
tenure_group	tenure	Bucket tenure into 0–12, 13–24, 25– 48, 49–72 months for clearer EDA	No (EDA only)
One-hot service/contract flags	Categorical columns	Convert nominal categories to numeric 0/1 indicators	Yes
Scaled numeric features	tenure, MonthlyCharges, TotalCharges	Standardise magnitude for Logistic Regression	Yes

Table 5. Engineered and Derived Features

11. MODEL DEVELOPMENT

11.1 Problem Formulation

The task is binary classification: predict whether a customer will churn (1) or stay (0). The input is the set of 30 encoded and scaled features; the target is the Churn indicator. This formulation matches the business need, which is to flag individual at-risk customers rather than to estimate a continuous quantity.

11.2 Model Selection

Two models were trained and compared. Logistic Regression was chosen as the primary model because it is well suited to binary classification, is fast and stable, and — importantly — produces interpretable coefficients that reveal which factors raise or lower the probability of churn. A Random Forest classifier was trained as a stronger, non-linear comparison and to obtain a second view of feature importance. Comparing a simple linear model with an ensemble model also allows a meaningful discussion of overfitting and generalisation.

11.3 Training the Model

Both models were trained on the 5,634-row training set using scikit-learn with a fixed random seed. The key training code is shown below; the full listing appears in Appendix A.

```

from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier

# Primary model: interpretable linear classifier
logreg = LogisticRegression(max_iter=1000, random_state=42)
logreg.fit(X_train, y_train)

# Comparison model: non-linear ensemble
rf = RandomForestClassifier(n_estimators=200, random_state=42)
rf.fit(X_train, y_train)

```

For Logistic Regression, `max_iter` was raised to 1,000 to ensure convergence on the 30-feature input. For the Random Forest, 200 trees were used to stabilise the ensemble. All other parameters were left at their scikit-learn defaults.

12. MODEL EVALUATION

Models were evaluated on the untouched 1,409-row test set. Because the classes are imbalanced, evaluation goes beyond accuracy to report precision, recall and F1-score for each class, together with the confusion matrix.

12.1 Classification Evaluation

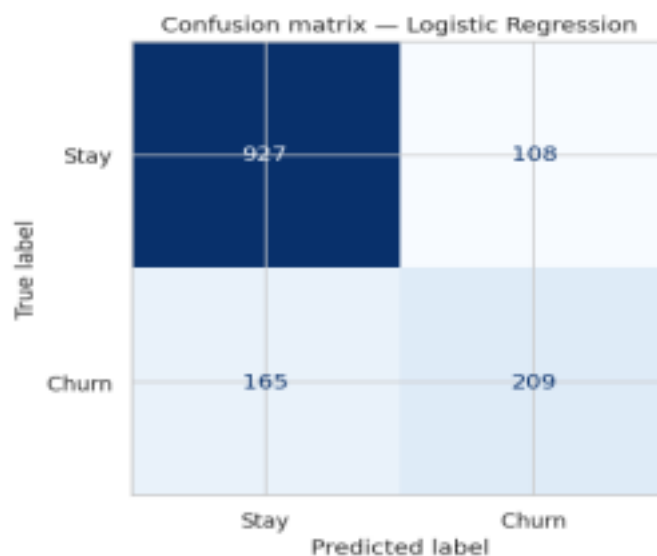


Figure 6. Confusion matrix — Logistic Regression (test set)

The Logistic Regression confusion matrix records 927 true negatives, 108 false positives, 165 false negatives and 209 true positives. In churn terms, a false negative (predicting a customer will stay when in fact they churn) is the costlier error, because it means an at-risk customer is missed and no retention action is taken. A false positive (flagging a loyal customer as at-risk) merely wastes a retention offer. The model currently produces more false negatives (165) than false positives (108), which is the aspect most worth improving. Table 6 reports the per-class metrics.

Class	Precision	Recall	F1-score	Support
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Stay (0)	0.849	0.896	0.872	1,035
Churn (1)	0.659	0.559	0.605	374
Accuracy			0.806	1,409
Macro average	0.754	0.727	0.738	1,409

Table 6. Classification Performance — Logistic Regression

Table 7 compares both models on headline metrics. Logistic Regression outperforms the Random Forest on every test-set measure while being far simpler and more interpretable.

Model	Test accuracy	Churn precision	Churn recall	Churn F1
Logistic Regression	0.806	0.659	0.559	0.605
Random Forest	0.794	0.642	0.508	0.567

Table 7. Classification Model Comparison (test set)

12.2 Overfitting and Underfitting

Comparing training and test performance is revealing. Logistic Regression scored 0.806 on training and 0.806 on test — essentially no gap — indicating a well-fitted model that generalises cleanly. The Random Forest, by contrast, scored 0.998 on training but only 0.794 on test, a gap of roughly 20 percentage points

that is a textbook symptom of overfitting: the ensemble has effectively memorised the training data. This comparison is the main reason Logistic Regression is preferred as the final model — it is not only simpler and more interpretable but also generalises better.

13. RESULTS AND FINDINGS

Bringing together the exploratory and modelling results, the key findings that answer the research questions are:

Finding 1 — Contract type is the dominant churn driver. Month-to-month customers churn at 42.7% versus 2.8% for two-year contracts (Figure 3), and “Contract: Two year” is the strongest churn-lowering coefficient in the model.

Finding 2 — Early tenure carries the highest risk. Churn falls monotonically from 47.4% in the first year to 9.5% beyond four years (Figure 4); tenure is one of the two largest churn-lowering coefficients.

Finding 3 — Fibre-optic internet and electronic-check payment are associated with elevated churn (41.9% and 45.3% respectively); fibre-optic service is the single largest churn-raising coefficient in the Logistic Regression model.

Finding 4 — The data are imbalanced (26.5% churn), so accuracy alone overstates performance (Figure 1). The chosen model achieves 80.6% accuracy but recall on churners of only 0.559, meaning it still misses a

substantial share of leavers.

Finding 5 — Logistic Regression is the best-performing and most reliable model, matching training and test accuracy exactly (0.806) while the Random Forest overfits badly (0.998 vs 0.794).

14. DISCUSSION

The results form a coherent and commercially intuitive picture. The variables that most strongly protect against churn — long contracts and long tenure — both represent forms of customer commitment, and the model confirms what the exploratory analysis suggested: customers who have signed up for longer, or who have already stayed longer, are far less likely to leave. The variables that raise churn tell a complementary story. Fibre-optic customers churn more despite (or because of) paying higher monthly charges, which may point to price sensitivity or to service-quality expectations that are not being met; electronic-check payers, who lack the automatic, low-friction renewal of card or bank-transfer methods, also churn more. These are observed associations, not proven causes, and should be read as such — for example, fibre-optic service may be a marker for a demanding, comparison-shopping customer segment rather than a direct cause of churn.

From a modelling standpoint, the most instructive outcome is the contrast between the two classifiers. The Random Forest's near-perfect training score is a caution rather than an achievement: it demonstrates how a flexible model can memorise noise and then fail to carry that performance to new data. Logistic Regression's stability, interpretability and marginally better test metrics make it the sensible choice for a decision-support tool, where being able to explain why a customer is flagged matters as much as the flag itself. The practitioner takeaway is that retention resources should be concentrated on the identifiable high risk segments, and that the model is useful for ranking risk even though its recall leaves room for improvement.

15. LIMITATIONS

- The dataset is a single cross-sectional snapshot of one telecom company; patterns may not generalise to other companies, markets or time periods.
- The analysis establishes associations, not causation — no controlled or longitudinal evidence supports causal claims about why customers churn.
- The classes are imbalanced (26.5% churn), and although stratification was used, the model's recall on churners (0.559) means a meaningful share of leavers is still missed.
- Potentially influential variables are absent, such as competitor pricing, network quality, customer support satisfaction and reasons for leaving.
- Only two model families were compared, both with largely default hyperparameters; more extensive tuning or alternative algorithms were outside the project scope.
- TotalCharges blanks for new customers were filled with zero; while defensible, this is an assumption about newly-joined customers rather than a measured value.

16. CONCLUSION

This project set out to understand and predict customer churn for a subscription telecom business. Through systematic cleaning — most notably repairing the TotalCharges field and resolving the blanks belonging to

new customers — followed by exploratory, statistical and predictive analysis, the study established that churn is concentrated among month-to-month, short-tenure, fibre-optic and electronic-check customers. A Logistic Regression classifier predicted churn on unseen data with 80.6% accuracy and generalised cleanly, outperforming a Random Forest that overfitted the training data. The objectives were met: the data were prepared, the drivers of churn were identified and quantified, predictive models were built and evaluated with appropriate metrics, and the findings were translated into recommendations. The overall project demonstrates a complete, reproducible data science workflow and produces insights that a retention team could act on directly.

17. RECOMMENDATIONS AND FUTURE WORK

17.1 Recommendations

Recommendation 1 — Because first-year, month-to-month customers show the highest churn (47.4% and 42.7% respectively), launch an early-lifecycle retention programme focused on the first three to six months, with proactive check-ins and onboarding support.

Recommendation 2 — Since two-year contracts churn at only 2.8%, offer targeted incentives — discounts or added services — to migrate high-risk month-to-month customers onto longer contracts.

Recommendation 3 — Given the elevated churn among fibre-optic subscribers (41.9%), investigate whether pricing or service quality is driving dissatisfaction in this premium segment and address the specific cause.

Recommendation 4 — As electronic-check payers churn at 45.3%, encourage a switch to automatic payment methods (card or bank transfer), which are associated with markedly lower churn.

Recommendation 5 — Use the Logistic Regression model to rank customers by churn probability and prioritise retention spend on the highest-risk individuals rather than applying offers uniformly.

17.2 Future Work

Future work could improve the model's recall on churners by addressing class imbalance directly — for example using class weighting, resampling techniques such as SMOTE, or tuning the decision threshold. Additional predictive power might come from richer data, including customer-support interactions, competitor pricing and satisfaction surveys. Alternative algorithms (gradient boosting, for instance) and systematic hyperparameter tuning could be evaluated, and the final model could be validated over multiple time periods before being deployed as a live retention-scoring system.

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APPENDIX A. CODE

IDRA Capstone 2026 — Customer Churn

Key code excerpts are shown below. The complete, runnable notebook is submitted alongside this report.

A.1 Data Loading and Inspection

```
import pandas as pd, numpy as np
df = pd.read_csv('P_4_WA_Fn-UseC_-Telco-Customer-Churn.csv')
print(df.shape) # (7043, 21)
df.info(); df['Churn'].value_counts(normalize=True)
```

A.2 Data Cleaning

```
df = df.drop(columns=['customerID'])
df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')
df['TotalCharges'] = df['TotalCharges'].fillna(0) # 11 new customers,
tenure=0 df['SeniorCitizen'] = df['SeniorCitizen'].map({0:'No', 1:'Yes'})
```

A.3 Preprocessing

```
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
df['Churn'] = df['Churn'].map({'No':0, 'Yes':1})
X = pd.get_dummies(df.drop(columns=['Churn']), drop_first=True)
y = df['Churn']
X_train,X_test,y_train,y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y)
num = ['tenure','MonthlyCharges','TotalCharges']
sc = StandardScaler()
X_train[num] = sc.fit_transform(X_train[num]) # fit on train only
X_test[num] = sc.transform(X_test[num])
```

A.4 EDA and Visualization

```
import seaborn as sns, matplotlib.pyplot as plt
rate = df.groupby('Contract')['Churn'].mean() # after mapping to 0/1
sns.barplot(x=rate.index, y=rate.values)
plt.title('Churn rate by contract type'); plt.show()
```

A.5 Model Training and Evaluation

```
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report, confusion_matrix
logreg = LogisticRegression(max_iter=1000, random_state=42)
logreg.fit(X_train, y_train)
pred = logreg.predict(X_test)
print(confusion_matrix(y_test, pred))
print(classification_report(y_test, pred))
```

APPENDIX B. ADDITIONAL FIGURES AND TABLES

The following figures support the interpretation but are not essential to the main narrative.

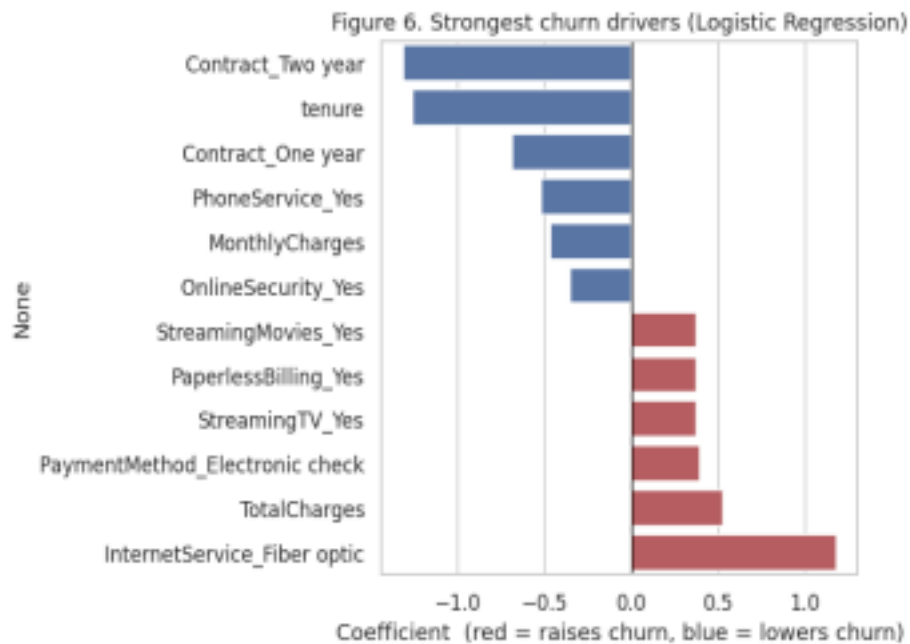


Figure 7. Strongest churn drivers (Logistic Regression coefficients). Positive values raise churn probability; negative values lower it.

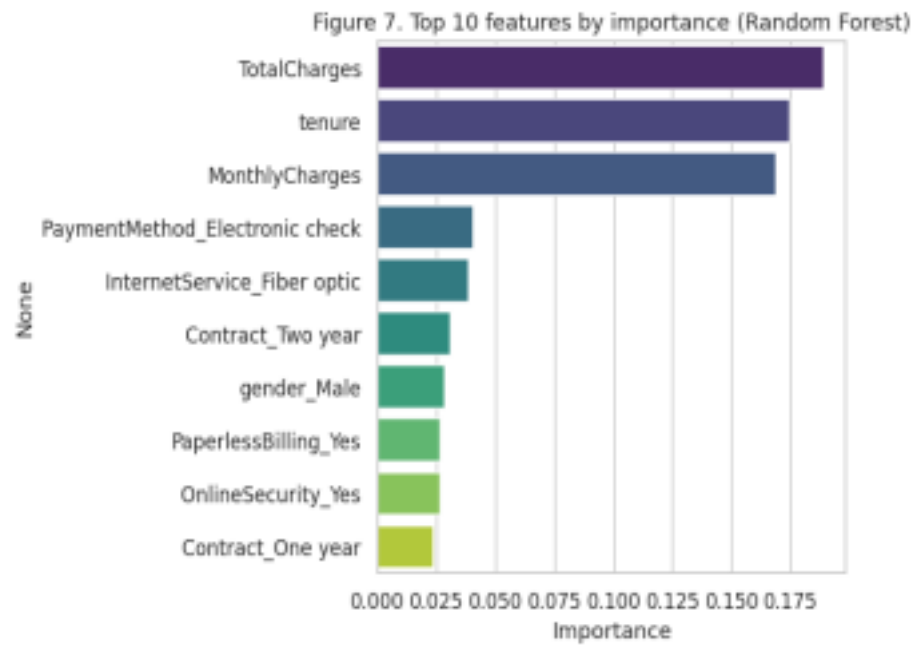


Figure 8. Top 10 features by importance (Random Forest).

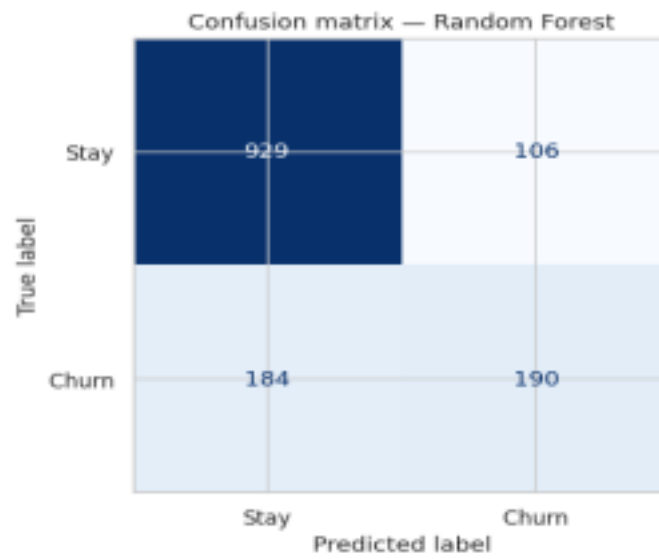


Figure 9. Confusion matrix — Random Forest (test set), shown for comparison.